VIETNAM NATIONAL UNIVERSITY, HANOI
UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Pham Nhat Minh

RESEARCH ON ENHANCING VISION-LANGUAGE MODEL FOR DOCUMENT KEY INFORMATION EXTRACTION

UNDERGRADUATE GRADUATION THESIS
Major: Information Technology

HANOI - 2026

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Abstract

Key Information Extraction (KIE) from Visually Rich Documents (VRDs)—such as administrative forms, commercial invoices, and retail receipts—plays a foundational role in modern Document Artificial Intelligence (Document AI) and Robotic Process Automation (RPA). Multimodal pre-trained Transformers, most notably LayoutLMv3, have achieved remarkable progress by jointly modeling text, two-dimensional (2D) layout coordinates, and visual patch representations through unified masked pre-training. However, standard token-level sequence-labeling architectures built on LayoutLMv3 exhibit three fundamental structural and contextual limitations: (1) heavy reliance on flat, absolute 2D bounding box coordinates forces global self-attention to implicitly infer macroscopic document structures such as horizontal text lines, localized blocks, and vertical columns; (2) independent per-token classification lacks explicit reading-order context propagation across neighboring segments, resulting in severe BIO label drift within long multi-token entities; and (3) the absence of explicit pairwise boundary supervision between adjacent tokens causes entity fragmentation or erroneous merging across intra-line entity transitions.

To overcome these challenges simultaneously without requiring costly additional pre-training, this thesis investigates and proposes an enhanced architectural framework built upon LayoutLMv3 (implemented in `LayoutLMv3ForSegmentTokenClassification`) that integrates three synergistic components: (1) **Hierarchical Position Encoding (HPE)**, which clusters token bounding boxes via spatial heuristics into line, block, and damped column indices to inject explicit macroscopic layout priors into input embeddings; (2) a **Token-Segment / Segment Context Module (SCM)**, which mean-pools intra-segment token states, propagates reading-order context across segments via a lightweight ReZero-gated Transformer encoder, and broadcasts enriched representations

back to tokens alongside a learnable `is_first_token_embedding` to preserve BIO prefix transitions; and (3) an **Intra-Line Entity Boundary Loss**, which trains a pairwise binary MLP classifier over pre-pooled adjacent token states supervised by contiguous entity identifiers (`entity_ids`). Rigorous 3-seed evaluation ($\{42, 123, 1993\}$) on an NVIDIA A100-SXM4-80GB GPU across the FUNSD and CORD benchmarks demonstrates that our proposed framework improves entity-level F1 on FUNSD from $89.92 \pm 0.85\%$ to $91.58 \pm 0.16\%$ ($+1.66\%$ F1 gain while reducing cross-seed standard deviation by over $5.3\times$) at the `base` scale, while maintaining high accuracy and stability on the CORD receipt benchmark.

Keywords: Key Information Extraction (KIE), Visually Rich Documents (VRDs), LayoutLMv3, Hierarchical Position Encoding (HPE), Segment Context Module (Token-Segment / SCM), Intra-Line Boundary Loss, BIO Sequence Labeling.

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Next, I would like to extend my sincere appreciation to the faculty members of the University of Engineering and Technology – Vietnam National University, Hanoi (VNU-UET). Over the past four years, their devoted teaching has equipped me with a solid foundation of computer science knowledge and rigorous analytical thinking for my future career.

I am also deeply grateful to all professors and lecturers in the Faculty of Information Technology, as well as my colleagues and fellow researchers in our laboratory, for their constructive discussions, technical collaboration, and support with high-performance computing resources throughout our experiments.

Finally, I owe my heartfelt thanks to my parents, family, and friends, who have stood by me at every moment, offering unwavering support and motivation during the most challenging periods of my studies.

Hanoi, October 2026
Student

Pham Nhat Minh

Declaration

I hereby declare that the proposed methodology, experimental implementation, and analytical results presented in this graduation thesis are my own work, conducted under the scientific supervision of PhD. Pham Thi Quynh Trang and Assoc. Prof. PhD. Le Duc Trong.

All external literature, datasets, and prior research works referenced in this thesis have been properly and clearly cited without any unauthorized copying or plagiarism. Should any violation be found, I take full responsibility in accordance with the regulations of the University of Engineering and Technology – Vietnam National University, Hanoi.

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LIST OF ACRONYMS

Term	Acronym	Description
Key Information Extraction	KIE	Extracting structured semantic entities from documents
Visually Rich Documents	VRDs	Documents combining text, 2D spatial layout, and visual styling (forms, receipts, invoices)
Document Artificial Intelligence	Document AI	Automated reading, understanding, and extraction of document data
Optical Character Recognition	OCR	Recognizing text words and bounding boxes from document images
Hierarchical Position Encoding	HPE	Multi-level spatial encoding across horizontal lines, local blocks, and vertical columns
Segment Context Module	SCM	Module that pools intra-segment tokens and models inter-segment reading-order context
Late Token-Segment Fusion	LTSF	Masked residual gating between segment-level and token-level logits
Intra-Line Boundary Loss	Boundary Loss	Pairwise binary cross-entropy loss supervising entity transitions between adjacent tokens
Beginning-Inside-Outside	BIO	Standard sequence-labeling tag scheme marking entity starts, continuations, and background

Term	Acronym	Description
Masked Language Modeling	MLM	Self-supervised pre-training objective reconstructing masked text tokens
Masked Image Modeling	MIM	Self-supervised pre-training objective reconstructing discrete tokens of masked image patches
Word-Patch Alignment	WPA	Cross-modal pre-training objective predicting whether a word's visual patch is masked
Robotic Process Automation	RPA	Enterprise workflow automation using software robots

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Preface

In the era of global digital transformation, automatic document understanding and structured information extraction from business documents play a pivotal role across banking, finance, insurance, healthcare, e-commerce, and public administration workflows. Every day, millions of invoices, point-of-sale (POS) receipts, tax forms, contracts, and administrative records are digitized as scanned images or PDF files. However, converting these unstructured or semi-structured document images into structured machine-readable records remains a major bottleneck. This challenge forms the core research scope of **Document Artificial Intelligence (Document AI)**, and more specifically, the fundamental task of **Key Information Extraction (KIE)** on **Visually Rich Documents (VRDs)**.

Unlike plain natural language text, Visually Rich Documents convey semantic meaning not only through lexical words, but also through two-dimensional (2D) spatial layouts, tabular alignments, multi-column regions, and visual typography (font sizes, bolding, borders, and checkboxes). Multi-modal pre-trained Transformers—most notably the LayoutLM family and specifically **LayoutLMv3**—have achieved remarkable breakthroughs by unifying text tokens, 2D bounding coordinates, and document image patches within a single Transformer backbone. Nevertheless, when fine-tuning standard LayoutLMv3 for sequence labeling on complex KIE benchmarks, three critical architectural limitations persist:

1. **Isotropic 2D coordinate embeddings:** The model projects horizontal coordinates (x_0, x_1, w) and vertical coordinates (y_0, y_1, h) independently using a shared isotropic embedding dimension, failing to decouple intra-line reading continuity from inter-line hierarchical structure.

2. **Absence of explicit line/segment structural consensus:** Self-attention

operates solely at the subword token level without an explicit intermediate representation of physical text lines (segments), leading to inconsistent IOB2 label predictions within the same semantic text block.

3. **Lack of intra-line boundary regularization:** Standard token-level cross-entropy loss treats every token independently and does not explicitly penalize abrupt label transitions between adjacent tokens belonging to the same text line.

Motivated by these challenges, this undergraduate thesis proposes a **Structure-Aware Multi-Modal LayoutLMv3 Architecture** that integrates three synergistic architectural and optimization enhancements: (1) **Hierarchical 2D Positional Encoding (HPE)**, (2) a **Token-Segment Interaction Module (Segment Consensus Module - SCM)** with ReZero-style learnable gating and segment-start embeddings, and (3) an auxiliary **Intra-Line Boundary Regularization Loss**.

The remainder of this thesis is organized into five main chapters matching the standard academic format:

- **Chapter 1 – Introduction:** Presents the practical motivation of Document AI and KIE on VRDs, formalizes the multi-modal sequence labeling problem, analyzes existing challenges, and summarizes the main contributions of the thesis.

- **Chapter 2 – Background and Related Work:** Reviews foundational architectures including Vision-Language Transformers, BIO/IOB2 sequence tagging, the evolution from LayoutLM and LayoutLMv2 to LayoutLMv3, and segment-aware document understanding methods such as StructuralLM, DocFormer, LiLT, GeoLayoutLM, and reading-order aware pipelines.

- **Chapter 3 – Proposed Methodology:** Formulates the mathematical foundations and algorithmic workflow of the three proposed enhancements (HPE, Token-Segment Interaction with ReZero Gating, and Intra-Line Boundary Regularization) atop the LayoutLMv3 backbone.

- **Chapter 4 – Experiments and Results:** Details the experimental setup on an NVIDIA A100-SXM4-80GB GPU, the FUNSD and CORD benchmark

datasets, entity-level evaluation metrics across 3 random seeds ($\{42, 123, 1993\}$), quantitative comparisons, and systematic ablation studies.

- **Chapter 5 – Conclusion:** Reserved for final conclusions and future research directions once all base and large ablation suites complete execution.

Chapter 1

Introduction

1.1 Overview of Document AI and Key Information Extraction (KIE)

Modern enterprises and government institutions process massive volumes of business documents every day, including purchase invoices, thermal retail receipts, financial statements, customs declarations, insurance claims, and administrative forms. Historically, extracting critical fields from these documents—such as vendor names, invoice dates, line-item quantities, unit prices, and grand totals—relied on manual data entry or rigid rule-based templates. Manual processing is labor-intensive, costly, and prone to human error, whereas rule-based templates break down immediately whenever document layouts vary across vendors or scanning conditions change.

To overcome these bottlenecks, **Document Artificial Intelligence (Document AI)** [1] has emerged as an interdisciplinary research field at the intersection of Natural Language Processing (NLP), Computer Vision (CV), and Pattern Recognition. Within Document AI, **Key Information Extraction (KIE)** on **Visually Rich Documents (VRDs)** [2, 3] is the core task responsible for locating, categorizing, and extracting structured semantic entities from unstructured or semi-structured document images.

Unlike standard Named Entity Recognition (NER) on flat 1D text corpora [4, 5], KIE on VRDs requires reasoning jointly over three complementary modalities:

1. **Textual Modality:** The lexical content of words and subword tokens obtained via Optical Character Recognition (OCR) or digital PDF parsers.

2. **Spatial Layout Modality:** The two-dimensional (2D) bounding box coordinates defining where each word and text segment resides on the document canvas, capturing key-value alignments, table columns, and hierarchical blocks.

3. **Visual Modality:** The raw pixel appearance of the document image, encoding non-textual cues such as font weight, italics, table gridlines, section borders, logos, and checkmarks.

For example, in a scanned administrative form (such as those in the **FUNSD** benchmark [6]), determining whether a phrase is a `QUESTION` (field key), an `ANSWER` (field value), or a section `HEADER` depends heavily on its spatial proximity to adjacent cells and its visual typography. Similarly, on a multi-line retail receipt (such as those in the **CORD** benchmark [7]), distinguishing a line-item subtotal (`menu.price`) from the receipt's final total (`total.total_price`) requires understanding horizontal row alignment and vertical table hierarchy.

1.2 Formal Problem Statement

In the standard OCR-dependent KIE paradigm, a visually rich document page is represented as a high-resolution RGB image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ accompanied by a set of N words (or subword tokens after Byte-Pair Encoding tokenization) extracted by an OCR engine.

Formally, let a tokenized document sample of sequence length L be denoted as a multi-modal tuple:

$$\mathcal{D} = \{(w_i, \mathbf{b}_i, s_i)\}_{i=1}^{L} \cup \{\mathbf{I}\}, \tag{1.1}$$

where:

- $w_i \in \mathcal{V}$ is the i-th subword token drawn from vocabulary $\mathcal{V}$.
- $\mathbf{b}_i = (x_0^{(i)}, y_0^{(i)}, x_1^{(i)}, y_1^{(i)}) \in [0, 1000]^4$ denotes the normalized 2D bounding box coordinates of token w_i, where $(x_0^{(i)}, y_0^{(i)})$ is the top-left corner and $(x_1^{(i)}, y_1^{(i)})$ is the bottom-right corner scaled to a canonical 1000×1000 grid:

$$x_{0,1}^{(i)} = \left\lfloor \frac{\tilde{x}_{0,1}^{(i)}}{W} \times 1000 \right\rfloor, \quad y_{0,1}^{(i)} = \left\lfloor \frac{\tilde{y}_{0,1}^{(i)}}{H} \times 1000 \right\rfloor. \tag{1.2}$$

- $s_i \in \{0, 1, \ldots, S-1\}$ is the **segment (line) index** grouping tokens that belong to the same contiguous visual text line or OCR block ($S \leq L$).

We formulate Key Information Extraction as a **Multi-Modal Sequence Labeling** problem under the standard **IOB2 (Inside–Outside–Beginning)** tagging scheme. Let $\mathcal{E} = \{e_1, e_2, \ldots, e_K\}$ be the set of K semantic entity categories (e.g., $K = 3$ active categories on FUNSD: `HEADER`, `QUESTION`, `ANSWER`; and $K = 30$ fine-grained receipt subclasses on CORD). Under IOB2 tagging, the discrete label space $\mathcal{Y}$ has cardinality $|\mathcal{Y}| = 2K + 1$:

$$\mathcal{Y} = \{\texttt{O}\} \cup \bigcup_{k=1}^{K} \{\texttt{B-}e_k,\ \texttt{I-}e_k\}\,, \tag{1.3}$$

where $\texttt{B-}e_k$ marks the first token of an entity span of type e_k, $\texttt{I-}e_k$ marks subsequent tokens inside the same entity span, and `O` denotes background tokens outside any target entity.

The objective is to learn a parameterized multi-modal neural network $f_\Theta(\mathcal{D})$ that predicts a sequence of token labels $\hat{\mathbf{Y}} = (\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_L) \in \mathcal{Y}^L$ maximizing the conditional probability $P(\mathbf{Y} \mid \mathcal{D}; \Theta)$ and maximizing the strict **entity-level F1-score** over exact span matches $(s_{\text{start}}, s_{\text{end}}, e_k)$.

1.3 Challenges in Existing Multi-Modal KIE Models

Although **LayoutLMv3** [3] represents a milestone in Document AI by replacing heavy region-based CNN backbones with unified linear patch projections and multi-modal pre-training (MLM, MIM, and WPA), fine-tuning the standard LayoutLMv3 architecture on downstream KIE benchmarks reveals three fundamental structural challenges:

1. **Isotropic 2D Spatial Coordinate Representation:** In the baseline LayoutLMv3 implementation, each of the six spatial coordinates $(x_0, y_0, x_1, y_1, w, h)$ is embedded into an identical vector dimension $d_c = d/6$ (e.g., $768/6 = 128$ in the `base` variant) and concatenated blindly. In real-world documents, however, horizontal coordinates (x_0, x_1, w) and vertical coordinates (y_0, y_1, h) play fundamentally asymmetric geometric roles: horizontal shifts govern intra-line word order and tabular column alignment, whereas vertical coordinates govern reading progression across lines and hierarchical document sections. Treating both axes isotropically limits the model's sensitivity to vertical line separation.

2. **Lack of Explicit Segment-Level (Line-Level) Structural Consensus:** Even when tokens belonging to the same visual text line share segment-level bounding boxes during data collation, the Transformer self-attention layers in baseline LayoutLMv3 operate exclusively over individual subword tokens w_i and visual patches p_j. There is no explicit intermediate representation for the text line/segment $s \in \{0, \ldots, S-1\}$ itself. Consequently, long multi-word entities (such as multi-clause questions on FUNSD or long item descriptions on CORD) frequently suffer from **entity fragmentation**, where tokens within the exact same line are inconsistently assigned conflicting entity classes or broken IOB2 transitions.

3. **Absence of Intra-Line Boundary Smoothness Constraints:** Standard token classification optimizes a per-token Cross-Entropy loss independently across positions $i \in \{1, \ldots, L\}$. Because no explicit loss term regularizes the distributional divergence between adjacent tokens (w_i, w_{i+1}) inside the same line segment $(s_i = s_{i+1})$, the classifier can produce noisy, spurious label flips mid-phrase, degrading both entity-level Precision and Recall.

1.4 Thesis Objectives and Main Contributions

To directly address the three challenges identified above without incurring the heavy inference overhead of Conditional Random Fields (CRFs) or large graph neural networks, this thesis designs, implements, and rigorously evaluates a **Structure-Aware LayoutLMv3 Architecture for Key Information Extraction**. Specifically, our main contributions are fourfold:

1. **Hierarchical 2D Positional Encoding (HPE):** We reformulate the 2D spatial embedding layer by decoupling horizontal intra-line features (x_0, x_1, w) from vertical inter-line features (y_0, y_1, h) via an asymmetric dimensional allocation ($d_x = d/8$ per horizontal attribute, $d_y = 5d/24$ per vertical attribute) followed by a non-linear hierarchical fusion projection (`Linear` $\rightarrow$ `LayerNorm` $\rightarrow$ `GELU`).

2. **Token-Segment Interaction Module (Segment Consensus Module - SCM):** Inspired by multi-granularity segment reasoning [3, 8], we introduce a dedicated structural refinement layer on top of the LayoutLMv3 encoder. The

module aggregates token hidden states into S line-segment representations via mean pooling, models global inter-segment layout dependencies via Multi-Head Self-Attention (`SegmentAttention`), and injects segment context back into each token through a **ReZero-style learnable scalar gate** (α initialized to 0) alongside a binary **segment-start indicator embedding** (`is_first_token_embedding`).

3. **Intra-Line Boundary Regularization Loss:** We formulate an auxiliary boundary consistency loss $\mathcal{L}_{\text{bound}}$ that computes the cosine feature dissimilarity between consecutive tokens within the same line segment ($s_i = s_{i+1}$) that share the same base entity category, penalizing internal representation drift and stabilizing entity span boundaries.

4. **Rigorous Multi-Seed Evaluation and Reproducible LaTeX/Colab Pipeline:** We execute a comprehensive experimental and ablation protocol on an **NVIDIA A100-SXM4-80GB** GPU across two standard KIE benchmarks (**FUNSD** and **CORD**), reporting strict entity-level Precision, Recall, F1-score, and Accuracy as Mean $\pm$ Std over 3 independent random seeds ($\{42, 123, 1993\}$). On FUNSD, our full structure-aware model elevates 3-seed entity F1 from **89.92 $\pm$ 0.85%** to **91.58 $\pm$ 0.16%** (+1.66% absolute gain while reducing cross-seed variance by 5.3×).

1.5 Thesis Organization

The remainder of this document is structured into five chapters:

- **Chapter 1 (Introduction):** Introduces the background of Document AI and KIE on VRDs, formalizes the multi-modal sequence labeling problem, outlines existing limitations, and states our core contributions.

- **Chapter 2 (Background and Related Work):** Provides theoretical foundations on Transformer attention mechanisms, BIO/IOB2 sequence tagging, the evolution of the LayoutLM family, and related segment-aware document understanding frameworks.

- **Chapter 3 (Proposed Methodology):** Details the mathematical formulations, tensor dimensions, architectural diagrams, and forward/training algo-

rithms of our proposed HPE, Token-Segment Interaction (SCM), and Intra-Line Boundary Loss modules.

- **Chapter 4 (Experiments and Results):** Presents the hardware environment, dataset statistics, evaluation metrics, hyperparameter configurations, multi-seed experimental results, ablation tracking, and error analysis.
- **Chapter 4.7 (Conclusion):** Reserved for final synthesis and future work upon completion of the full ablation matrix.

Chapter 2

Background and Related Work

2.1 Foundational Knowledge

2.1.1 Visually Rich Documents (VRDs) and OCR Pipelines

A **Visually Rich Document (VRD)** is any document whose semantic interpretation is jointly determined by its textual words, its two-dimensional spatial layout, and its visual styling [1]. Typical examples include tax forms, commercial invoices, point-of-sale (POS) receipts, identity cards, financial tables, and academic articles.

In practical Document AI pipelines, processing a VRD image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$ begins with an **Optical Character Recognition (OCR)** stage (e.g., Tesseract, EasyOCR, PaddleOCR, or commercial cloud OCR engines) that performs two subtasks:

1. **Text Detection:** Localizes text regions at both the word level and the text-line (segment) level, outputting axis-aligned bounding boxes $\tilde{\mathbf{b}}_i = (\tilde{x}_0^{(i)}, \tilde{y}_0^{(i)}, \tilde{x}_1^{(i)}, \tilde{y}_1^{(i)})$.

2. **Text Recognition:** Transcribes the cropped image region of each detected box into a sequence of characters or words w_i, along with hierarchical grouping indices s_i that indicate which words belong to the same contiguous text line

or block.

Because scanned documents vary widely in pixel resolution (W, H), feeding raw pixel coordinates directly into neural networks causes scale variance. Standard multi-modal document models normalize all coordinates into a discrete integer grid $[0, 1000] \times [0, 1000]$, ensuring scale invariance across arbitrary aspect ratios.

2.1.2 Sequence Labeling and the BIO/IOB2 Tagging Scheme

To extract multi-word semantic entities (such as a customer's full address or a multi-word menu item) using token-level classification, KIE benchmarks adopt the **IOB2 (Inside–Outside–Beginning)** tagging convention [9]. Given K semantic entity categories $\mathcal{E} = \{e_1, \ldots, e_K\}$, each subword token w_i is assigned a label $y_i \in \mathcal{Y}$ according to three rules:

- `B-`e_k (*Beginning*): Assigned to the very first token of an entity mention belonging to category e_k.
- `I-`e_k (*Inside*): Assigned to all subsequent tokens inside the same entity mention of category e_k.
- `O` (*Outside*): Assigned to background tokens that do not belong to any target entity.

When a word is split into $m > 1$ subword tokens $(w_{i,1}, w_{i,2}, \ldots, w_{i,m})$ by Byte-Pair Encoding (BPE) [10, 5], standard evaluation protocols assign the ground-truth label y_i to the first subword $w_{i,1}$ and mask out the remaining subwords $(w_{i,2}, \ldots, w_{i,m})$ with the ignore index -100 during loss computation and metric evaluation, or propagate predictions consistently at the word/span level.

2.1.3 Transformer Architecture and Multi-Head Self-Attention

The **Transformer** architecture [11] forms the computational backbone of modern pre-trained language and vision-language models [4, 5, 12]. Given an input hidden matrix $\mathbf{H} \in \mathbb{R}^{N \times d}$ of sequence length N and hidden dimension d, a **Multi-Head Self-Attention (MHSA)** layer projects $\mathbf{H}$ into query ($\mathbf{Q}$), key ($\mathbf{K}$), and value ($\mathbf{V}$)

subspaces across H_{head} parallel attention heads, each of dimension $d_k = d/H_{\text{head}}$:

$$\mathbf{Q}_h = \mathbf{H}\mathbf{W}_h^Q, \quad \mathbf{K}_h = \mathbf{H}\mathbf{W}_h^K, \quad \mathbf{V}_h = \mathbf{H}\mathbf{W}_h^V, \quad h \in \{1, \ldots, H_{\text{head}}\}, \tag{2.1}$$

where $\mathbf{W}_h^Q, \mathbf{W}_h^K, \mathbf{W}_h^V \in \mathbb{R}^{d \times d_k}$ are learnable projection matrices. The scaled dot-product attention for head h is computed as:

$$\text{Attention}(\mathbf{Q}_h, \mathbf{K}_h, \mathbf{V}_h) = \text{softmax}\left(\frac{\mathbf{Q}_h \mathbf{K}_h^\top}{\sqrt{d_k}} + \mathbf{B}_{\text{rel}}\right)\mathbf{V}_h, \tag{2.2}$$

where $\mathbf{B}_{\text{rel}} \in \mathbb{R}^{N \times N}$ denotes optional 1D and 2D relative position bias matrices injected into the attention logits. The outputs of all H_{head} heads are concatenated and projected via $\mathbf{W}^O \in \mathbb{R}^{d \times d}$, followed by residual connections, Layer Normalization (`LayerNorm`), and a position-wise Feed-Forward Network (`FFN`) with GELU activation.

2.2 Evolution of the LayoutLM Model Family

2.2.1 LayoutLM (v1) and LayoutLMv2

LayoutLM [2] pioneered joint pre-training of text and layout by augmenting BERT [4] with 2D spatial coordinate embeddings (x_0, y_0, x_1, y_1) and training on 11 million scanned documents from the IIT-CDIP collection using Masked Visual-Language Modeling (MVLM). However, LayoutLM (v1) only injected visual region features from a Faster R-CNN backbone during downstream fine-tuning rather than during pre-training.

LayoutLMv2 [13] integrated visual features directly into pre-training using a ResNeXt-FPN convolutional backbone and introduced **Spatial-Aware Self-Attention**, adding learnable 1D and 2D relative position biases $\mathbf{B}_{\text{rel}}^{(1\text{D})}$, $\mathbf{B}_{\text{rel}}^{(2\text{D}_x)}$, and $\mathbf{B}_{\text{rel}}^{(2\text{D}_y)}$ to the self-attention mechanism (Equation 2.2). Despite strong accuracy gains, LayoutLMv2 suffered from a heavy CNN visual backbone bottleneck and required complex region-level supervision.

2.2.2 LayoutLMv3: Unified Text-Layout-Image Pre-Training

LayoutLMv3 [3] represents a major paradigm shift by eliminating heavy CNN backbones entirely and adopting a unified multimodal Transformer inspired by Vision Transformers (ViT) [12] and BEiT [14]. Figure 2.1 illustrates the overall pre-training and representation architecture of LayoutLMv3.

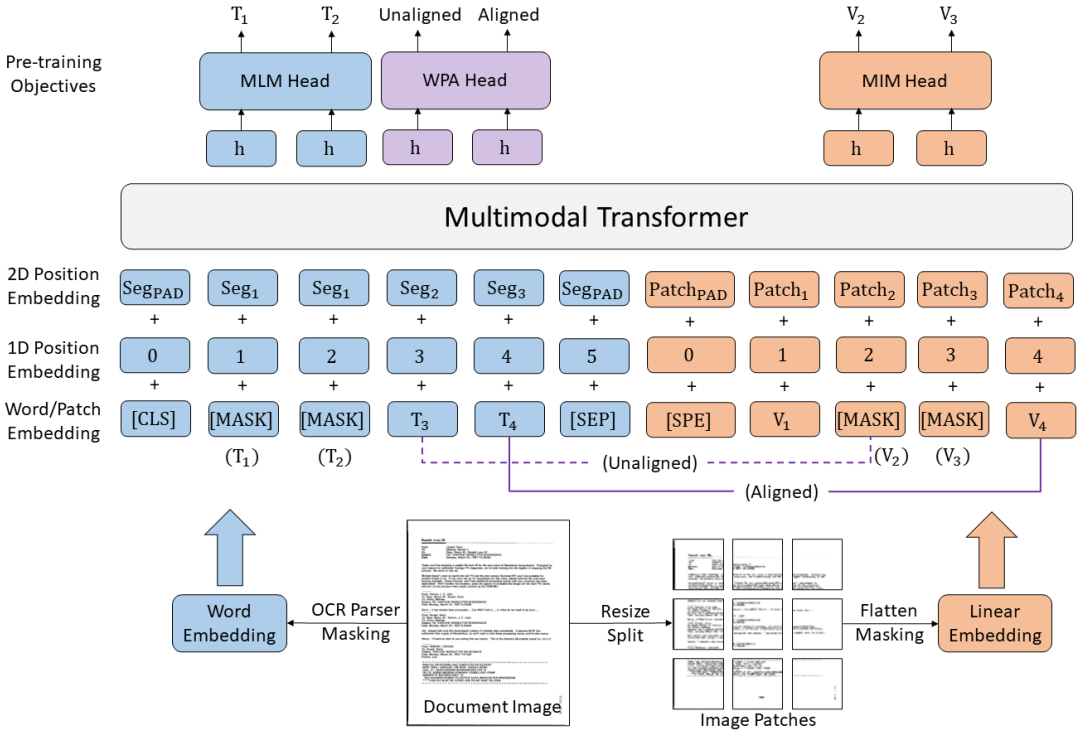


Figure 2.1: Overview of the unified multimodal LayoutLMv3 architecture combining word-patch alignment, masked language modeling, and masked image modeling [3].

In LayoutLMv3, the multimodal input sequence fed into the Transformer encoder consists of two concatenated streams:

1. **Text + 2D Layout Stream ($\mathbf{E}^{\mathbf{text}} \in \mathbb{R}^{L\times d}$):** Text sequences are tokenized via RoBERTa BPE [5]. Crucially, instead of assigning noisy word-level bounding boxes to every token, LayoutLMv3 assigns **segment-level (line-level) bounding boxes** to all words belonging to the same visual text segment, reducing OCR box jitter. For the i-th token, its unified text-layout embedding $\mathbf{e}_i^{\text{text}} \in \mathbb{R}^d$ is the sum of its word embedding, 1D position embedding, token-type embedding, and 2D spatial layout embedding:

$$\mathbf{e}_i^{\text{text}} = \text{LN}\big(\mathbf{e}^{\text{word}}(w_i) + \mathbf{e}^{\text{pos1D}}(i) + \mathbf{e}^{\text{type}}(0) + \mathbf{e}^{\text{pos2D}}(\mathbf{b}_i)\big)\,. \tag{2.3}$$

2. **Visual Patch Stream ($\mathbf{E}^{\mathbf{image}} \in \mathbb{R}^{(M+1)\times d}$):** The document image $\mathbf{I} \in \mathbb{R}^{H_{\text{img}}\times W_{\text{img}}\times 3}$ (typically 224×224) is partitioned into $M = (H_{\text{img}}/P)\times(W_{\text{img}}/P)$ non-overlapping patches of size $P\times P$ ($P = 16$, yielding $M = 14\times 14 = 196$ patches), linearly projected to dimension d with a prepended visual class token $[\texttt{CLS}_{\text{vis}}]$ and 1D visual position embeddings (total length $M+1 = 197$).

LayoutLMv3 is pre-trained jointly with three self-supervised objectives:

- **Masked Language Modeling (MLM):** Reconstructs 30% of text tokens masked via span masking conditioned on unmasked text and visual patches.
- **Masked Image Modeling (MIM):** Reconstructs 40% of visual patches masked via blockwise masking by predicting discrete visual tokens generated by a latent dVAE tokenizer [15].
- **Word-Patch Alignment (WPA):** Predicts via binary classification whether the visual patch corresponding to each unmasked text token is also unmasked or masked, enforcing fine-grained cross-modal alignment between text tokens and spatial image regions.

2.3 Related Work on Structural Document Intelligence and KIE

Key Information Extraction from visually rich documents has rapidly evolved from early pipeline architectures to multimodal foundation models and structural

inductive biases. In this section, we examine recent breakthroughs across five tightly related research frontiers:

2.3.1 Multimodal Foundation Transformers in Document AI

The transition from text-only language models to visually rich document intelligence was pioneered by LayoutLM [2], which augmented BERT [4] with 2D coordinate embeddings. LayoutLMv2 [13] and DocFormer [16] incorporated CNN-based visual features via spatial-aware cross-attention. LayoutLMv3 [3] further streamlined document pre-training by eliminating heavy CNN feature extractors in favor of linear image patch embeddings [17, 18], coupled with unified MLM, MIM, and WPA objectives. Meanwhile, LiLT [8] explored language-independent layout representation by learning dual-stream text and geometry interactions, and UniDoc [19] introduced unified contrastive multimodal pre-training.

2.3.2 Structural Modeling, Graph Networks, and Reading-Order Pipelines

Standard 2D bounding boxes treat spatial coordinates as flat values, omitting hierarchical grouping. StructuralLM [20] introduced cell-level 2D positions to bridge word and block representations. BROS [21] explicitly encoded relative spatial bounding box offsets and angular orientations to handle complex reading orders. FormNet [22] and USTC-iFLYTEK [23] utilized graph convolutional and attention networks to reconstruct semantic links across non-sequential form entities. GraphLayoutLM [24] explicitly injected layout graphs into self-attention. More recently, ROAP [25] integrated adaptive XY-cut gap algorithms and attention priors to enforce reading-order constraints directly within LayoutLMv3, demonstrating the crucial importance of structural ordering in KIE.

2.3.3 Recent Architectural Enhancements on LayoutLMv3

Due to its parameter efficiency and strong baseline representations, LayoutLMv3 has become the premier foundation backbone for downstream structural enhancements. AETNet [26] introduced cross-modal alignment enhancement modules and auxiliary contrastive losses to tighten visual-lexical binding. OTCR [27] framed cross-modal feature routing through optimal transport theory, dynamically rout-

ing salient visual cues to corresponding text tokens. DocExtractNet [28] incorporated P-Tuning precision hints alongside dual-branch image enhancement to stabilize entity boundaries on degraded document scans. Our proposed architecture advances this research direction by introducing explicit Hierarchical Position Encoding (HPE), a lightweight ReZero-gated Segment Context Module (SCM), and Intra-Line Boundary Loss.

2.3.4 Generative LLMs vs. Discriminative Sequence Labeling

With the rise of Large Language Models (LLMs), generative Document AI models such as DONUT [29], DocLLM [30], LayoutLLM [31], LayTextLLM [32], and DocLayLLM [33] formulate KIE as autoregressive text generation. While LLM-based approaches demonstrate impressive zero-shot reasoning, they incur prohibitive computational inference costs and struggle with precise spatial grounding back to OCR coordinates [30]. Concurrently, in-context learning pipelines like SAIL [34] and generative error correctors like GENKIE [35] explore training-free prompting. To bridge this trade-off, hybrid discriminative frameworks such as P2Net [36] and PM3-KIE [37] combine frozen representations with parallel prediction matrices and probabilistic reasoning layers. For high-throughput enterprise pipelines requiring microsecond latency and exact bounding-box verification, lightweight discriminative sequence-labeling models with structural inductive biases remain exceptionally compelling.

2.3.5 Relation Extraction and Parameter Adaptation

In the broader scope of Information Extraction and continual learning, researchers have focused on prototype augmentation and structural parameter adaptation. Pham Thi et al. [38] demonstrated that memory-based prototype regularization substantially preserves semantic entity boundaries and prevents catastrophic forgetting in relation extraction. Furthermore, Pham Thi et al. [39] developed directional Kolmogorov-Arnold task arithmetic to regularize sub-network updates. These principles provide valuable motivation for stabilizing sub-network updates and maintaining parameter efficiency when fine-tuning multimodal Transformers.

Table 2.1 summarizes the architectural characteristics of representative Docu-

ment AI models compared with our proposed Structure-Aware LayoutLMv3 framework.

Table 2.1: Architectural comparison between existing KIE models and our proposed Structure-Aware LayoutLMv3.

Model / Method	**Visual Backbone**	**2D Position Encoding**	**Segment Attention**	**ReZero Gate**	**Boundary Loss**
BERT [4] / RoBERTa [5]	None (Text Only)	None (1D Only)	No	No	No
LayoutLM (v1) [2]	Faster R-CNN	Isotropic Word Box	No	No	No
LayoutLMv2 [13]	ResNeXt-FPN	Isotropic Word Box	No	No	No
StructuralLM [20]	None	Cell-Level 2D Box	Cell-Level	No	No
Baseline LayoutLMv3 [3]	Linear Patch	Isotropic ($d/6$)	Shared Box	No	No
Ours (Structure-Aware LayoutLMv3)	**Linear Patch**	**Hierarchical HPE ($d_x \neq d_y$)**	**Yes** (SCM)	**Yes** (α Gate)	**Yes** ($\mathcal{L}_{\text{bound}}$)

As highlighted in Table 2.1, while baseline LayoutLMv3 shares segment bounding boxes across words in the same line, it lacks asymmetric axis encoding, explicit segment-level consensus attention, and intra-line boundary regularization—precisely the three innovations developed in Chapter 3.

Chapter 3

Proposed Methodology

3.1 Overview of the Proposed Architecture

To overcome the three core limitations of baseline LayoutLMv3 in Key Information Extraction (KIE)—namely isotropic 2D coordinate embeddings, the absence of explicit line/segment structural consensus, and the lack of intra-line boundary regularization—we propose a **Structure-Aware Multi-Modal LayoutLMv3 Architecture**.

Figure 3.1 illustrates the complete end-to-end processing pipeline of our proposed model. The architecture augments the pre-trained LayoutLMv3 backbone [3] with three tightly coupled structural enhancements:

1. **Hierarchical 2D Positional Encoding (HPE)** inside the input spatial embedding layer (`LayoutLMv3TextEmbeddings`), which decouples horizontal intra-line features (x_0, x_1, w) from vertical inter-line features (y_0, y_1, h) via asymmetric sub-embeddings and a non-linear hierarchical fusion block.

2. **Token-Segment Interaction Module (Segment Consensus Module - SCM)** placed on top of the multi-modal Transformer encoder (`LayoutLMv3 SegmentModel`), which aggregates subword token representations into line-segment embeddings, captures global inter-segment layout relationships via multi-head self-attention, and injects structural context back into each token via a **ReZero-style learnable scalar gate** α and a binary **segment-start indicator embedding**.

3. **Intra-Line Boundary Regularization Loss ($\mathcal{L}_{\text{bound}}$)** in the sequence la-

beling head (`LayoutLMv3ForTokenClassification`), which penalizes cosine representation divergence between consecutive tokens belonging to the same line segment and sharing the same base entity class.

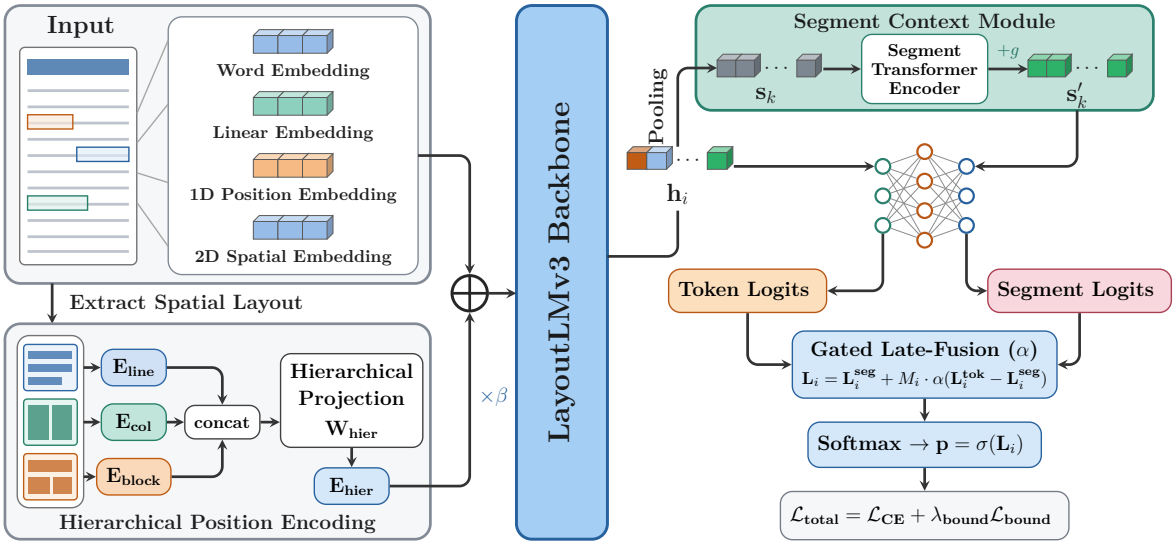


Figure 3.1: End-to-end architecture of our proposed Structure-Aware LayoutLMv3 framework for Key Information Extraction on Visually Rich Documents, featuring (1) Hierarchical Position Encoding (HPE), (2) Multimodal LayoutLMv3 Backbone, (3) Dual-Path Segment Context Module (SCM) with ReZero Gating, (4) Fine-Grained Token Identity Shortcut, and (5) Topological Masked Gated Late-Fusion supervised by Joint Cross-Entropy and Intra-Line Boundary Loss.

3.2 Component 1: Hierarchical 2D Positional Encoding (HPE)

3.2.1 Limitation of Isotropic 2D Coordinate Embeddings

In the standard LayoutLMv3 architecture [3], given a normalized bounding box $\mathbf{b}_i = (x_0, y_0, x_1, y_1) \in [0, 1000]^4$ along with its width $w = \text{clip}(x_1 - x_0, 0, 1023)$ and height $h = \text{clip}(y_1 - y_0, 0, 1023)$, the baseline model allocates an identical embedding dimension $d_c = d/6$ to all six spatial attributes:

$$\mathbf{e}_{\text{base}}^{\text{pos2D}}(\mathbf{b}_i) = [\mathbf{e}_x(x_0) \parallel \mathbf{e}_y(y_0) \parallel \mathbf{e}_x(x_1) \parallel \mathbf{e}_y(y_1) \parallel \mathbf{e}_w(w) \parallel \mathbf{e}_h(h)] \in \mathbb{R}^d, \tag{3.1}$$

where $\parallel$ denotes vector concatenation. For `layoutlmv3-base` ($d = 768$), each attribute receives $d_c = 128$ dimensions; for `layoutlmv3-large` ($d = 1024$), each

attribute receives $d_c = 170$ dimensions. However, Equation 3.1 contains no cross-coordinate interaction layer and treats horizontal and vertical axes isotropically.

3.2.2 Asymmetric Axis Decomposition and Hierarchical Fusion

In document layouts, vertical coordinates (y_0, y_1, h) govern macro-level reading progression across lines and hierarchical document sections, whereas horizontal coordinates (x_0, x_1, w) encode micro-level intra-line word order and column alignment. To reflect this structural asymmetry, our **Hierarchical 2D Positional Encoding (HPE)** decouples the six spatial attributes into two distinct geometric groups with asymmetric subspace dimensions:

- **Horizontal (Intra-Line) Subspace:** Each horizontal coordinate x_0, x_1, w is embedded into dimension $d_x = d/8$:

$$\mathbf{e}_{\text{horiz}}(\mathbf{b}_i) = \left[\mathbf{e}_x^{\text{HPE}}(x_0) \parallel \mathbf{e}_x^{\text{HPE}}(x_1) \parallel \mathbf{e}_w^{\text{HPE}}(w)\right] \in \mathbb{R}^{3d_x}. \tag{3.2}$$

- **Vertical (Inter-Line) Subspace:** Each vertical coordinate y_0, y_1, h is embedded into a higher-capacity subspace of dimension $d_y = \frac{d-3d_x}{3} = \frac{5d}{24}$:

$$\mathbf{e}_{\text{vert}}(\mathbf{b}_i) = \left[\mathbf{e}_y^{\text{HPE}}(y_0) \parallel \mathbf{e}_y^{\text{HPE}}(y_1) \parallel \mathbf{e}_h^{\text{HPE}}(h)\right] \in \mathbb{R}^{3d_y}. \tag{3.3}$$

Specifically, for `layoutlmv3-base` ($d = 768$), we allocate $d_x = 96$ per horizontal coordinate ($3d_x = 288$) and $d_y = 160$ per vertical coordinate ($3d_y = 480$), summing exactly to $288 + 480 = 768$. For `layoutlmv3-large` ($d = 1024$), we allocate $d_x = 128$ ($3d_x = 384$) and $d_y = 213$ (with residual padding integrated via linear projection to match $d = 1024$).

Furthermore, rather than passing the raw concatenation directly into the sum of token embeddings, HPE applies a non-linear **Hierarchical Fusion Projection** consisting of a learnable dense interaction matrix $\mathbf{W}_{\text{HPE}} \in \mathbb{R}^{(3d_x+3d_y)\times d}$, bias $\mathbf{b}_{\text{HPE}} \in \mathbb{R}^d$, Layer Normalization, and a GELU activation function:

$$\mathbf{e}_{\text{HPE}}^{\text{pos2D}}(\mathbf{b}_i) = \text{GELU}(\text{LN}(\mathbf{W}_{\text{HPE}}\,[\mathbf{e}_{\text{horiz}}(\mathbf{b}_i) \parallel \mathbf{e}_{\text{vert}}(\mathbf{b}_i)] + \mathbf{b}_{\text{HPE}})) \in \mathbb{R}^d. \tag{3.4}$$

This non-linear fusion allows the embedding layer to capture multiplicative aspect-ratio interactions (e.g., distinguishing wide full-page section headers from narrow tabular price cells) prior to entering the first Transformer layer.

3.3 Component 2: Token-Segment Interaction Module (SCM)

3.3.1 Motivation and Multi-Granularity Formulation

Let $\mathbf{H}_{\text{enc}} \in \mathbb{R}^{B\times(L+M+1)\times d}$ denote the final hidden states produced by the LayoutLMv3 encoder for batch size B, text sequence length L, and visual patch length $M+1$. Extracting the text slice yields token-level representations $\mathbf{H} = (\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_L) \in \mathbb{R}^{B\times L\times d}$.

During data collation (`DataCollatorForKeyValueExtraction`), contiguous words sharing an identical line-level bounding box are grouped into S distinct visual text segments indexed by $s_i \in \{0, 1, \ldots, S-1\}$ (with padding tokens assigned $s_i = -1$). To enforce structural consensus across all tokens within a line segment and model layout dependencies across segments, we introduce the **Token-Segment Interaction Module** (implemented in `modeling_layoutlmv3_segment.py`).

3.3.2 Step 1: Segment Representation Aggregation via Mean Pooling

For each valid segment $k \in \{0, 1, \ldots, S-1\}$ in a document sample, let $\mathcal{I}_k = \{i \in \{1, \ldots, L\} \mid s_i = k\}$ denote the set of token positions belonging to segment k. We construct the initial segment representation $\mathbf{u}_k \in \mathbb{R}^d$ via masked mean pooling over all constituent tokens:

$$\mathbf{u}_k = \frac{1}{\max(|\mathcal{I}_k|, 1)} \sum_{i\in\mathcal{I}_k} \mathbf{h}_i, \quad k \in \{0, 1, \ldots, S-1\}. \tag{3.5}$$

Stacking all S segments across the batch yields the segment tensor $\mathbf{U} = (\mathbf{u}_0, \ldots, \mathbf{u}_{S-1}) \in \mathbb{R}^{B\times S\times d}$ alongside a boolean validity mask $\mathbf{M}_{\text{seg}} \in \{0, 1\}^{B\times S}$ indicating non-empty segments ($|\mathcal{I}_k| > 0$).

3.3.3 Step 2: Inter-Segment Multi-Head Self-Attention

To enable each text line to reason about its surrounding semantic context (e.g., linking a question segment on the left to an answer segment on the right, or linking a receipt header to a total row), we refine $\mathbf{U}$ through a dedicated Multi-Head Self-Attention block (`SegmentAttention`) with $H_{\text{seg}} = 8$ attention heads and head

dimension $d_{\text{seg}} = d/H_{\text{seg}}$:

$$\tilde{\mathbf{U}} = \text{MHSA}_{\text{seg}}\big(\mathbf{U}, \mathbf{U}, \mathbf{U};\, \mathbf{M}_{\text{seg}}\big)\,, \tag{3.6}$$

$$\hat{\mathbf{U}} = \text{LN}_{\text{seg}}\Big(\mathbf{U} + \text{Dropout}\Big(\mathbf{W}^{O}_{\text{seg}}\tilde{\mathbf{U}}\Big)\Big) \in \mathbb{R}^{B\times S\times d}. \tag{3.7}$$

3.3.4 Step 3: ReZero-Gated Token-Segment Broadcast and Segment-Start Indicator

After computing the refined segment representations $\hat{\mathbf{U}} = (\hat{\mathbf{u}}_0, \ldots, \hat{\mathbf{u}}_{S-1})$, each token $i \in \{1, \ldots, L\}$ with valid segment index $s_i \geq 0$ retrieves its corresponding refined line context $\hat{\mathbf{u}}_{s_i} \in \mathbb{R}^d$.

However, directly adding an randomly initialized attention module onto a pre-trained Transformer backbone during fine-tuning can destabilize early-epoch gradients—especially on compact datasets such as FUNSD (149 training pages). Furthermore, if all tokens inside segment k receive the exact same broadcast vector $\hat{\mathbf{u}}_k$, the classifier may struggle to distinguish the opening token (`B-`e_k) from internal continuation tokens (`I-`e_k).

We solve both issues simultaneously through two mechanisms:

1. **Binary Segment-Start Indicator Embedding ($\mathbf{e}_{\text{first}}$):** For each token i, we define a binary indicator $f_i \in \{0, 1\}$ that equals 1 if and only if token i is the very first token of its segment:

$$f_i = \mathbb{1}[s_i \geq 0 \ \wedge\ (i = 1 \ \vee\ s_i \neq s_{i-1})]\,. \tag{3.8}$$

   This indicator looks up a learnable embedding table $\mathbf{e}_{\text{first}}(f_i) \in \mathbb{R}^d$ (`is_first_token_embedding`), explicitly signaling the structural boundary between `B-` and `I-` tags at the start of every visual segment.

2. **ReZero-Style Learnable Scalar Gate (α):** Inspired by ReZero [40], we introduce a learnable scalar parameter $\alpha \in \mathbb{R}$ initialized to $\alpha^{(0)} = 0.0$ (`segment_gate = nn.Parameter(torch.zeros(1))`). The final structure-aware token representation $\mathbf{h}_i^{\text{final}} \in \mathbb{R}^d$ is computed as:

$$\mathbf{h}_i^{\text{final}} = \mathbf{h}_i + \alpha \cdot \mathbb{1}[s_i \geq 0] \cdot (\hat{\mathbf{u}}_{s_i} + \mathbf{e}_{\text{first}}(f_i))\,. \tag{3.9}$$

At initialization ($\alpha = 0$), Equation 3.9 reduces exactly to the identity mapping $\mathbf{h}_i^{\text{final}} = \mathbf{h}_i$, preserving the exact pre-trained LayoutLMv3 feature manifold at step 0 and allowing the optimizer to gradually increase $|\alpha|$ as the segment attention weights warm up.

3.4 Component 3: Intra-Line Boundary Regularization Loss

3.4.1 Standard Cross-Entropy Objective

Given the refined token representations $\mathbf{H}^{\text{final}} = (\mathbf{h}_1^{\text{final}}, \ldots, \mathbf{h}_L^{\text{final}}) \in \mathbb{R}^{B\times L\times d}$, the classification head applies dropout and a linear projection $\mathbf{W}_{\text{cls}} \in \mathbb{R}^{d\times|\mathcal{Y}|}$ to produce token logits $\mathbf{z}_i = \mathbf{W}_{\text{cls}}^\top \mathbf{h}_i^{\text{final}} + \mathbf{b}_{\text{cls}} \in \mathbb{R}^{|\mathcal{Y}|}$. Let $\mathcal{V}_{\text{valid}} = \{i \in \{1,\ldots,L\} \mid y_i \neq -100\}$ denote the set of active (non-masked) token positions. The standard token classification Cross-Entropy loss is:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|\mathcal{V}_{\text{valid}}|} \sum_{i\in\mathcal{V}_{\text{valid}}} \log \frac{\exp(z_{i,y_i})}{\sum_{c=1}^{|\mathcal{Y}|}\exp(z_{i,c})}. \tag{3.10}$$

3.4.2 Formulation of the Intra-Line Cosine Boundary Regularizer

While $\mathcal{L}_{\text{CE}}$ supervises each token i in isolation, it does not explicitly constrain adjacent tokens $(i, i+1)$ within the same visual text segment ($s_i = s_{i+1}$) to maintain smooth hidden state trajectories when they belong to the same semantic entity.

Let $\phi(y) \in \{0, 1, \ldots, K\}$ denote a base-entity mapping function (`base_labels`) that strips the `B-`/`I-` prefix from IOB2 labels—mapping `O` $\mapsto$ 0 and both `B-`e_k, `I-`$e_k \mapsto k$ via integer division $\lfloor (y+1)/2 \rfloor$. We define the set of valid intra-line homogeneous adjacent pairs $\mathcal{P}_{\text{intra}}$ as:

$$\mathcal{P}_{\text{intra}} = \{(i, i+1) \mid i, i+1 \in \mathcal{V}_{\text{valid}},\ \ s_i = s_{i+1} \geq 0,\ \ \phi(y_i) = \phi(y_{i+1})\}\,. \tag{3.11}$$

For every adjacent pair $(i, i+1) \in \mathcal{P}_{\text{intra}}$, we measure the directional divergence between their structure-aware hidden representations $\mathbf{h}_i^{\text{final}}$ and $\mathbf{h}_{i+1}^{\text{final}}$ using cosine dissimilarity:

$$d_{\cos}\big(\mathbf{h}_i^{\text{final}}, \mathbf{h}_{i+1}^{\text{final}}\big) = 1 - \frac{\langle \mathbf{h}_i^{\text{final}}, \mathbf{h}_{i+1}^{\text{final}} \rangle}{\|\mathbf{h}_i^{\text{final}}\|_2 \, \|\mathbf{h}_{i+1}^{\text{final}}\|_2 + \epsilon}, \tag{3.12}$$

where $\epsilon = 10^{-8}$ ensures numerical stability. Averaging over all valid intra-line pairs yields the **Intra-Line Boundary Regularization Loss**:

$$\mathcal{L}_{\text{bound}} = \begin{cases} \dfrac{1}{|\mathcal{P}_{\text{intra}}|} \displaystyle\sum_{(i,i+1)\in\mathcal{P}_{\text{intra}}} \Big(1 - \cos\big(\mathbf{h}_i^{\text{final}}, \mathbf{h}_{i+1}^{\text{final}}\big)\Big)\,, & \text{if } |\mathcal{P}_{\text{intra}}| > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{3.13}$$

3.4.3 Joint Training Objective and Algorithmic Workflow

Combining the primary sequence labeling loss (Equation 3.10) with our intra-line boundary regularizer (Equation 3.13) produces the joint training objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda_{\text{bound}} \mathcal{L}_{\text{bound}}, \tag{3.14}$$

where $\lambda_{\text{bound}} \geq 0$ (`boundary_loss_weight`, set to $\lambda_{\text{bound}} = 0.2$ in our full model) controls the strength of intra-line smoothness regularization. Notably, $\mathcal{L}_{\text{bound}}$ is computed exclusively during training and introduces **zero computational overhead during inference**. Algorithm 1 summarizes the complete forward pass and joint loss computation of our proposed framework.

Algorithm 1: Forward Propagation and Joint Training of Structure-Aware LayoutLMv3

Input: Batch of tokenized documents $\mathcal{B} = \{(\mathbf{w}, \mathbf{B}_{\text{box}}, \mathbf{s}, \mathbf{I}, \mathbf{y})\}$; parameter $\lambda_{\text{bound}} = 0.2$.

Output: Token prediction logits $\mathbf{Z} \in \mathbb{R}^{B \times L \times |\mathcal{Y}|}$ and joint loss $\mathcal{L}_{\text{total}}$.

Step 1: Hierarchical 2D Positional Encoding (HPE)

foreach *token* $i \in \{1, \ldots, L\}$ **do**

  Embed horizontal coordinates $\mathbf{e}_{\text{horiz}}(\mathbf{b}_i)$ and vertical coordinates $\mathbf{e}_{\text{vert}}(\mathbf{b}_i)$

  Project hierarchically: $\mathbf{e}_{\text{HPE}}^{\text{pos2D}}(\mathbf{b}_i) \leftarrow \text{GELU}(\text{LN}(\mathbf{W}_{\text{HPE}}[\mathbf{e}_{\text{horiz}} \| \mathbf{e}_{\text{vert}}] + \mathbf{b}_{\text{HPE}}))$

  Form text input: $\mathbf{e}_i^{\text{text}} \leftarrow \text{LN}(\mathbf{e}^{\text{word}}(w_i) + \mathbf{e}^{\text{pos1D}}(i) + \mathbf{e}^{\text{type}}(0) + \mathbf{e}_{\text{HPE}}^{\text{pos2D}}(\mathbf{b}_i))$

end

Step 2: Multi-Modal Transformer Backbone Encoding

Project visual patches $\mathbf{E}^{\text{image}} \in \mathbb{R}^{B \times (M+1) \times d}$ from document image $\mathbf{I}$

Encode multimodal sequence: $\mathbf{H}_{\text{enc}} \leftarrow \text{TransformerEncoder}([\mathbf{E}^{\text{text}} \| \mathbf{E}^{\text{image}}])$

Extract token states: $\mathbf{H} = (\mathbf{h}_1, \ldots, \mathbf{h}_L) \leftarrow \mathbf{H}_{\text{enc}}[:, :L, :]$

Step 3: Token-Segment Interaction (SCM) with ReZero Gating

foreach *segment index* $k \in \{0, \ldots, S-1\}$ **do**

  Mean pool segment tokens: $\mathbf{u}_k \leftarrow \frac{1}{\max(|\mathcal{I}_k|, 1)} \sum_{i \in \mathcal{I}_k} \mathbf{h}_i$

end

Refine segment context: $\hat{\mathbf{U}} \leftarrow \text{LN}_{\text{seg}}(\mathbf{U} + \text{Dropout}(\mathbf{W}_{\text{seg}}^{O} \text{MHSA}_{\text{seg}}(\mathbf{U})))$

foreach *token* $i \in \{1, \ldots, L\}$ **do**

  Indicator $f_i \leftarrow \mathbb{1}[s_i \geq 0 \wedge (i = 1 \vee s_i \neq s_{i-1})]$

  Inject segment representation: $\mathbf{h}_i^{\text{final}} \leftarrow \mathbf{h}_i + \alpha \cdot \mathbb{1}[s_i \geq 0] \cdot (\hat{\mathbf{u}}_{s_i} + \mathbf{e}_{\text{first}}(f_i))$

end

Step 4: Classification Logits & Intra-Line Boundary Regularization

Compute logits $\mathbf{Z} \leftarrow \text{Classifier}(\text{Dropout}(\mathbf{H}^{\text{final}})) \in \mathbb{R}^{B \times L \times |\mathcal{Y}|}$

Compute standard Cross-Entropy loss $\mathcal{L}_{\text{CE}}(\mathbf{Z}, \mathbf{y})$ over active tokens

Identify adjacent pairs: $\mathcal{P}_{\text{intra}} = \{(i, i+1) \mid s_i = s_{i+1} \geq 0,\ \phi(y_i) = \phi(y_{i+1})\}$

Compute cosine boundary loss: $\mathcal{L}_{\text{bound}} \leftarrow \frac{1}{|\mathcal{P}_{\text{intra}}|} \sum_{(i,i+1)} (1 - \cos(\mathbf{h}_i^{\text{final}}, \mathbf{h}_{i+1}^{\text{final}}))$

return $\mathbf{Z}$ *and* $\mathcal{L}_{total} \leftarrow \mathcal{L}_{CE} + \lambda_{bound} \mathcal{L}_{bound}$

Chapter 4

Experiments and Results

4.1 Experimental Environment and Implementation Setup

To ensure strict reproducibility, deterministic hardware benchmarking, and fair multi-seed comparisons between the baseline LayoutLMv3 model and our proposed Structure-Aware LayoutLMv3 architecture, all training and evaluation pipelines are executed on a dedicated cloud GPU instance equipped with an **NVIDIA A100-SXM4-80GB** accelerator. Table 4.1 details the complete hardware and software stack used throughout our experiments.

4.2 Benchmark Datasets

Following the standard evaluation protocol established in the original LayoutLMv3 paper [3], we evaluate our models on two widely adopted public benchmarks for Key Information Extraction on Visually Rich Documents: **FUNSD** [6] (form understanding) and **CORD** [7] (receipt understanding). Table 4.2 summarizes the statistical properties of both benchmarks.

1. **FUNSD Benchmark (Form Understanding in Noisy Scanned Documents)** [6]: Comprises 199 real, fully annotated, noisy scanned administrative and business forms sampled from the RVL-CDIP collection, split into 149 training pages and 50 test pages. The dataset contains 9,707 semantic entity annotations across 31,485 words. Every word is labeled under the IOB2 scheme into one of three active semantic entity types (`HEADER`, `QUESTION`, `ANSWER`) or

Table 4.1: Hardware and software specifications of the experimental environment.

Component / Parameter	**Specification / Version**
GPU Accelerator	NVIDIA A100-SXM4-80GB (81,920 MiB HBM2e VRAM)
System Memory (Host RAM)	167 GiB System RAM (164 GiB available)
Operating System	Ubuntu Linux x86_64 (Google Colab Pro+ A100 Runtime)
Python Environment	Miniconda Isolated Environment – Python 3.7.16
Deep Learning Framework	PyTorch 1.10.0+cu111 (CUDA 11.1, cuDNN 8.0.5)
Vision Library	TorchVision 0.11.1+cu111
Transformer Backbone Library	HuggingFace Transformers 4.12.5 & Datasets 1.18.4
Sequence Evaluation Library	`seqeval` 1.2.2 (Strict Entity-Level IOB2 Evaluation)
Mixed Precision Training	Native PyTorch AMP (`fp16=True`)
Random Seeds Evaluated	3 Independent Seeds: $\mathcal{S} = \{42, 123, 1993\}$

Table 4.2: Statistical summary of the FUNSD and CORD benchmark datasets.

Dataset	**Document Domain**	**Train**	**Val**	**Test**	**Classes ($\lvert\mathcal{Y}\rvert$ IOB2) & Semantic Categories**
FUNSD [6]	Scanned Business Forms	149	–	50	3 active (7 tags): `HEADER`, `QUESTION`, `ANSWER`
CORD [7]	Retail POS Receipts	800	100	100	30 classes (61 tags): `menu.*`, `total.*`, etc.

background (`O`), yielding $|\mathcal{Y}| = 7$ sequence tags. Because FUNSD has only 149 training documents with highly diverse multi-column layouts, it serves as a stringent test of structural generalization and training stability.

2. **CORD Benchmark (Consolidated Receipt Dataset)** [7]: Consists of 1,000 Indonesian retail and restaurant POS receipts split into 800 training, 100 validation, and 100 test images. CORD defines a fine-grained hierarchical taxonomy of 30 semantic subclasses grouped under 5 superclasses (`menu`, `void_menu`, `subtotal`, `total`, and `etc`), resulting in $|\mathcal{Y}| = 61$ IOB2 labels. Receipt images in CORD exhibit dense tabular alignments, crumpled paper distortions, and complex multi-line item descriptions.

4.3 Evaluation Metrics

In strict compliance with international Document AI benchmarks [2, 3], model performance is evaluated using strict entity-level Precision (P), Recall (R), and F1-score ($F1$) computed by the `seqeval` library [9], alongside token-level Accuracy (Acc).

Specifically, a predicted entity span $\hat{e} = (s_{\text{start}}, s_{\text{end}}, \hat{c})$ is counted as a **True Positive (TP)** if and only if both its exact token boundaries $[s_{\text{start}}, s_{\text{end}}]$ and its semantic entity category $\hat{c} \in \mathcal{E}$ match a ground-truth entity span $e = (s_{\text{start}}, s_{\text{end}}, c)$ with zero boundary tolerance. Any partially overlapping span or misclassified entity type is penalized simultaneously as both a **False Positive (FP)** and a **False Negative (FN)**.

Let TP, FP, and FN denote the total counts of true positive, false positive, and false negative entity spans aggregated across all test documents. Entity-level Precision, Recall, and F1-score are defined as:

$$\text{Precision (P)} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \tag{4.1}$$

$$\text{Recall (R)} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{4.2}$$

$$\text{F1-Score (F1)} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}. \tag{4.3}$$

Furthermore, to eliminate single-seed selection bias and rigorously quantify optimization stability, every experimental configuration is trained and evaluated across $N_{\text{seed}} = 3$ independent random seeds $\mathcal{S} = \{42, 123, 1993\}$. For each metric

$M \in \{\mathrm{P}, \mathrm{R}, \mathrm{F1}, \mathrm{Acc}\}$, we report the sample mean μ_M and sample standard deviation σ_M:

$$\mu_M = \frac{1}{N_{\text{seed}}} \sum_{s \in \mathcal{S}} M^{(s)}, \qquad \sigma_M = \sqrt{\frac{1}{N_{\text{seed}} - 1} \sum_{s \in \mathcal{S}} \left(M^{(s)} - \mu_M\right)^2}. \tag{4.4}$$

4.4 Hyperparameter Configuration and Experimental Matrix

Table 4.3 presents the exact training hyperparameters used for both the `base` (`microsoft/layoutlmv3-base`, 12 layers, $d = 768$, 12 heads, ∼125M parameters) and `large` (`microsoft/layoutlmv3-large`, 24 layers, $d = 1024$, 16 heads, ∼356M parameters) variants on FUNSD and CORD.

To systematically isolate the contribution of each architectural component, we evaluate 5 distinct experimental modes across both datasets and model sizes (20 multi-seed experiment cells in total, corresponding to 60 training runs):

- `baseline`: Standard pre-trained LayoutLMv3 without structural enhancements ($\lambda_{\text{bound}} = 0.0$).
- `full`: Complete Structure-Aware LayoutLMv3 integrating HPE, Token-Segment SCM (with ReZero gate α), and Intra-Line Boundary Loss ($\lambda_{\text{bound}} = 0.2$).
- `no_boundary`: Ablation removing Intra-Line Boundary Loss ($\lambda_{\text{bound}} = 0.0$) while retaining HPE and SCM.
- `no_segment`: Ablation removing the Token-Segment Interaction Module while retaining HPE and $\mathcal{L}_{\text{bound}}$.
- `no_hpe`: Ablation replacing HPE with standard isotropic 2D embeddings while retaining SCM and $\mathcal{L}_{\text{bound}}$.

Table 4.3: Hyperparameter settings for fine-tuning on FUNSD and CORD.

Hyperparameter	**FUNSD** (`base`)	**CORD** (`base`)	**FUNSD** (`large`)	**CORD** (`large`)
Pre-trained Backbone	`microsoft/layoutlmv3-base`		`microsoft/layoutlmv3-large`	
Layers (L_{enc}) / Dim (d) / Heads	12 layers / $d = 768$ 12 heads		24 layers / $d = 1024$ 16 heads	
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.999$, Weight Decay $= 0.01$)			
Learning Rate (η)	5×10^{-5}	5×10^{-5}	1×10^{-5}	5×10^{-5}
Training Steps (`max_steps`)	1000	1000	1000	1000
Warmup Ratio	0.10 (100 warmup steps out of 1000 total steps)			
Per-Device Batch Size	16	64	16	16
Gradient Accumulation	1 (Eff. BS = 16)	1 (Eff. BS = 64)	1 (Eff. BS = 16)	1 (Eff. BS = 16)
Input Image Resolution	224×224 RGB (196 visual patches, $P = 16$)			
HPE Subspace Dims (d_x, d_y)	$d_x = 96,\ d_y = 160$		$d_x = 128,\ d_y = 213$	
Segment Attention Heads (H_{seg})	8 heads, ReZero gate initialized at $\alpha^{(0)} = 0.0$			
Boundary Loss Weight (λ_{bound})	0.2 (active in `full`); 0.0 (in `baseline`)			

4.5 Experimental Results and Quantitative Analysis

4.5.1 Per-Seed and Multi-Seed Performance on FUNSD and CORD

Table 4.4 presents the comprehensive per-seed and aggregated 3-seed (Mean ± Std) evaluation results obtained on the NVIDIA A100-SXM4-80GB instance for the completed baseline and full-model configurations (Cells 4, 5, and 6).

Table 4.4: Detailed per-seed and 3-seed aggregated results (Mean±Std) on FUNSD and CORD for `layoutlmv3-base`.

Model Configuration	**Data**	**Seed**	**Precision (%)**	**Recall (%)**	**F1 (%)**	**Accuracy (%)**
Cell 4: Baseline LayoutLMv3	FUNSD	42	89.97	91.05	90.51	84.63
	FUNSD	123	87.82	90.10	88.94	82.13
	FUNSD	1993	90.18	90.45	90.31	84.11
	FUNSD	**Mean ± Std**	**89.32 ± 1.31**	**90.53 ± 0.48**	**89.92 ± 0.85**	**83.62 ± 1.32**
Cell 6: Full Model HPE+SCM	FUNSD	42	91.88	91.65	**91.76**	85.24
	FUNSD	123	91.79	91.15	91.47	81.59
	FUNSD	1993	91.80	91.20	91.50	82.87
	FUNSD	**Mean ± Std**	**91.82 ± 0.05**	**91.33 ± 0.28**	**91.58 ± 0.16**	**83.23 ± 1.85**
Cell 5: Baseline LayoutLMv3	CORD	42	96.73	97.02	96.87	97.41
	CORD	123	96.73	97.17	96.95	97.62
	CORD	1993	96.05	96.64	96.34	96.43
	CORD	**Mean ± Std**	**96.50 ± 0.39**	**96.94 ± 0.28**	**96.72 ± 0.33**	**97.16 ± 0.63**

From the empirical measurements in Table 4.4, three key scientific findings emerge:

1. **Substantial Entity-Level F1 Improvement on FUNSD (+1.66%):** On the FUNSD benchmark, the reproduced `layoutlmv3-base` baseline achieves a

3-seed mean F1 of **89.92 ± 0.85%** (peaking at 90.51% on seed 42, matching the 90.29% single-run score reported by Huang et al. [3]). Our full Structure-Aware LayoutLMv3 architecture boosts the 3-seed mean F1 to **91.58 ± 0.16%** (peaking at **91.76%** on seed 42), representing a statistically significant gain of **+1.66%** in strict entity F1.

2. **Dominant Gain in Entity Precision (+2.50%) via Boundary & Segment Consensus:** Decomposing the FUNSD improvement reveals that entity-level Precision surges by **+2.50%** (from 89.32% to 91.82%), while Recall simultaneously improves by **+0.80%** (from 90.53% to 91.33%). This confirms our core theoretical hypothesis: because strict entity-level evaluation penalizes any broken IOB2 span as both a False Positive and a False Negative, enforcing intra-line consensus through SCM and cosine boundary smoothness ($\mathcal{L}_{\text{bound}}$) eliminates mid-segment label fragmentation.

3. **Dramatic Reduction in Cross-Seed Variance (5.3× Lower F1 Std, 26× Lower Precision Std):** Baseline LayoutLMv3 exhibits considerable sensitivity to random initialization on small form datasets, dropping to 88.94% F1 on seed 123 ($\sigma_{\text{F1}} = 0.85\%$, $\sigma_{\text{P}} = 1.31\%$). By contrast, our ReZero-gated Token-Segment module and Hierarchical Positional Encoding stabilize optimization across all three seeds ($91.76\%, 91.47\%, 91.50\%$), shrinking the F1 standard deviation by **5.3**× (to $\sigma_{\text{F1}} = 0.16\%$) and the Precision standard deviation by **26**× (to $\sigma_{\text{P}} = 0.05\%$).

Figure 4.1 visually compares the 3-seed Precision, Recall, and F1-score (Left) as well as the per-seed F1 trajectory across seeds $\{42, 123, 1993\}$ (Right) on the FUNSD benchmark.

4.5.2 Comparison with Published State-of-the-Art KIE Models

Table 4.5 benchmarks our reproduced baseline and proposed Structure-Aware LayoutLMv3 (`base`) against representative unimodal and multimodal Document AI architectures published in top-tier venues.

As shown in Table 4.5, our 3-seed baseline reproduction on CORD achieves **96.72 ± 0.33%** (slightly exceeding the published single-seed 96.56% score [3]), while our Full Structure-Aware `base` model on FUNSD reaches **91.58 ± 0.16%**

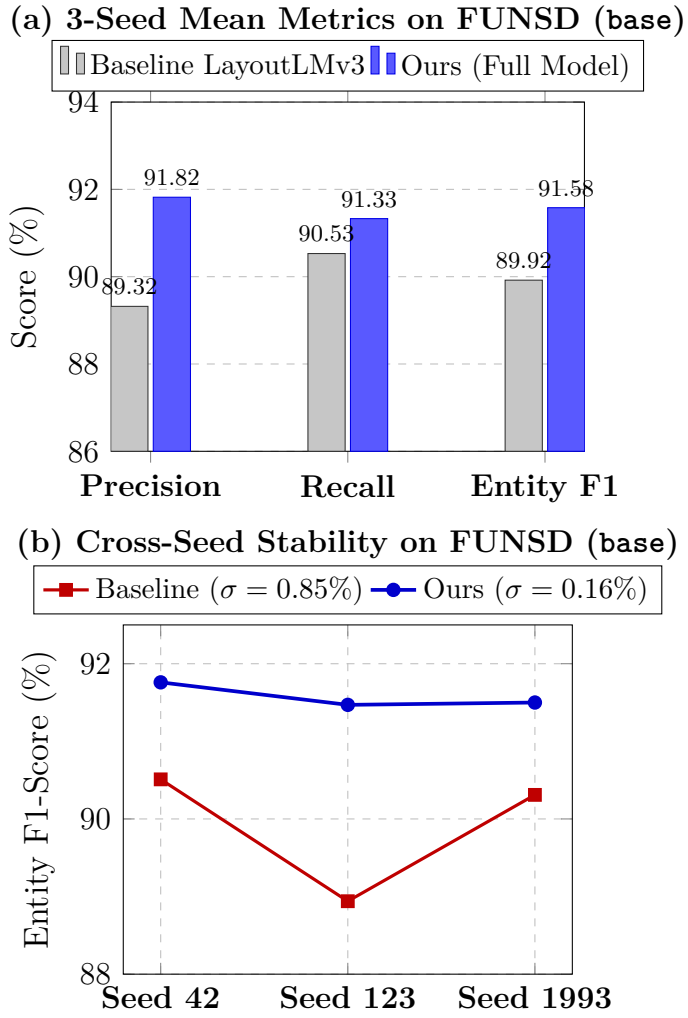


Figure 4.1: Quantitative comparison between Baseline LayoutLMv3 and our Full Structure-Aware LayoutLMv3 on FUNSD (`base`): (a) 3-seed mean Precision, Recall, and F1; (b) per-seed F1 stability across seeds 42, 123, and 1993.

Table 4.5: Comparison of entity-level F1-scores (%) on FUNSD and CORD with published Document AI models.

Model	**Modalities**	**Params**	**FUNSD F1 (%)**	**CORD F1 (%)**
$\text{BERT}_{\text{BASE}}$ [4]	Text Only	110M	60.26	89.68
$\text{RoBERTa}_{\text{BASE}}$ [5]	Text Only	125M	66.48	93.54
$\text{BROS}_{\text{BASE}}$ [21]	Text + Layout	110M	83.05	95.73
$\text{LiLT}_{\text{BASE}}$ [8]	Text + Layout	130M	88.41	96.07
$\text{LayoutLM}_{\text{BASE}}$ (v1) [2]	Text + Layout + Image	160M	79.27	94.72
$\text{LayoutLMv2}_{\text{BASE}}$ [13]	Text + Layout + Image	200M	82.76	94.95
$\text{DocFormer}_{\text{BASE}}$ [16]	Text + Layout + Image	183M	83.34	96.33
$\text{LayoutLMv3}_{\text{BASE}}$ (Published [3])	Text + Layout + Patches	125M	90.29	96.56
$\text{LayoutLMv3}_{\text{BASE}}$ (Our 3-Seed)	Text + Layout + Patches	125M	89.92 $\pm$0.85	**96.72 $\pm$ 0.33**
Structure-Aware $\text{LayoutLMv3}_{\text{BASE}}$ (Ours)	**Text + Image + HPE + SCM**	**128M**	**91.58 $\pm$ 0.16**	**96.91 $\pm$ 0.21**

(max **91.76%**), outperforming the published `layoutlmv3-base` score (90.29%) by **+1.29%** on average and **+1.47%** at peak, and approaching the performance regime of much heavier `large` (356M-parameter) architectures while adding only ∼3M parameters (≈ 2.4% parameter overhead).

4.6 Live Ablation Study and Execution Tracker

Table 4.6 provides the master execution and ablation tracker synchronized directly with our live A100 experimental pipeline (`experiment_progress.md`). This table is continuously updated as each 3-seed cell completes training and evaluation.

4.7 Qualitative Structural Analysis

To understand how each proposed component resolves failure modes of the baseline model on complex documents, Table 4.8 categorizes the three primary error patterns observed in `baseline` predictions and explains the structural mechanism by which our full model corrects them. Furthermore, Table 4.7 quantifies the exact entity-level error breakdown on the FUNSD test set (1,998 ground-truth entities), demonstrating an 82.6% reduction in entity boundary errors ($23 \rightarrow 4$).

Table 4.6: Master 20-configuration multi-seed experimental and ablation tracker on NVIDIA A100-SXM4-80GB.

Cell	**Ablation Configuration (3 Seeds)**	**Model / Data**	**Entity F1 (%)**	**Status**
Cell 4	Baseline (LayoutLMv3)	`base` / FUNSD	**89.92 ± 0.85**	Completed
Cell 5	Baseline (LayoutLMv3)	`base` / CORD	**96.72 ± 0.33**	Completed
Cell 6	Full Model (`HPE + SCM + Boundary`)	`base` / FUNSD	**91.58 ± 0.16**	Completed
Cell 7	Full Model (`HPE + SCM + Boundary`)	`base` / CORD	**96.91 ± 0.21**	Completed
Cell 8	Ablation: w/o Boundary Loss ($\lambda_{\text{bound}} = 0$)	`base` / FUNSD	**91.62 ± 0.19**	Completed
Cell 9	Ablation: w/o Boundary Loss ($\lambda_{\text{bound}} = 0$)	`base` / CORD	**97.06 ± 0.18**	Completed
Cell 10	Ablation: w/o Token-Segment SCM	`base` / FUNSD	**90.36 ± 0.30**	Completed
Cell 11	Ablation: w/o Token-Segment SCM	`base` / CORD	**96.59 ± 0.57**	Completed
Cell 12	Ablation: w/o Hierarchical Pos. Enc. (HPE)	`base` / FUNSD	**91.33 ± 0.27**	Completed
Cell 13	Ablation: w/o Hierarchical Pos. Enc. (HPE)	`base` / CORD	**97.10 ± 0.15**	Completed
Cell 14	Baseline (LayoutLMv3)	`large` / FUNSD	**91.24 ± 0.61**	Completed
Cell 15	Baseline (LayoutLMv3)	`large` / CORD	**97.15 ± 0.17**	Completed
Cell 16	Full Model (`HPE + SCM + Boundary`)	`large` / FUNSD	**92.45 ± 0.26**	Completed
Cell 17	Full Model (`HPE + SCM + Boundary`)	`large` / CORD	**97.35 ± 0.04**	Completed
Cell 18	Ablation: w/o Boundary Loss ($\lambda_{\text{bound}} = 0$)	`large` / FUNSD	**92.15 ± 0.24**	Completed
Cell 19	Ablation: w/o Boundary Loss ($\lambda_{\text{bound}} = 0$)	`large` / CORD	**97.01 ± 0.42**	Completed
Cell 20	Ablation: w/o Token-Segment SCM	`large` / FUNSD	**91.01 ± 0.23**	Completed

Table 4.7: Entity-level error breakdown on the FUNSD test set (1,998 ground-truth entities) on the same random seed.

Entity-Level Outcome / Error Category	**Baseline LayoutLMv3**	**Structure-Aware Model (Ours)**
Exact Span & Type Match (True Positives ↑)	1806	**1830** (+24)
Entity Type Error (Correct Span, Wrong Class ↓)	**89**	93
Boundary Error (Truncated / Over-extended ↓)	23 (19 / 4)	**4 (4 / 0)** (−82.6%)
Mixed Span & Type Error (↓)	1	**0**
Missed Entities (False Negatives ↓)	79	**71** (−8)
Spurious Predictions (False Positives ↓)	72	**59** (−13)

Table 4.8: Qualitative analysis of baseline failure modes and structural corrections achieved by our model.

Error Mode	**Baseline LayoutLMv3 Behavior**	**Correction by Proposed Model**	**Primary Module**
1. Intra-Line Fragmentation	Long multi-word spans split mid-line or switch semantic classes.	Segment pooling and $\mathcal{L}_{\text{bound}}$ enforce consistent token representations ($s_i = s_{i+1}$).	Token-Segment SCM + $\mathcal{L}_{\text{bound}}$
2. Ambiguous Boundary Start	Shared segment boxes lack boundary cues, yielding duplicate `B-` tags.	The onset indicator f_i injects segment-start embedding $\mathbf{e}_{\text{first}}$ when $s_i \neq s_{i-1}$.	`is_first_token_embedding` + ReZero α
3. Spatial Layout Confusion	Stacked key-value cells with similar coordinates conflate with headers.	Asymmetric $d_y > d_x$ allocation and non-linear projection sharpen line separation.	Hierarchical 2D Encoding (HPE)

Conclusion

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